

Project 1

Site: [Eduvos LMS](#)

Course: Front-end and Backend Technologies and Frameworks Assessments

Book: Project 1

Printed by: Lesedi Chuene

Date: Wednesday, 29 April 2026, 11:52 AM

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1. Assignment

Faculty:	Information Technology
Module Code:	ITFFA4-14
Module Name:	Front-End and Back-End Technologies and Frameworks
Content Writer:	Innocent Manganyi
Internal Moderation:	Mokgapi London Mashabela
Copy Editor:	Kyle Keens
Total Marks:	100

Submission Week:

Block 2 Week 7

This module is presented on NQF level 8

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [\[1\]](#)

This is an individual assignment

This assignment contributes 20% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

2. Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted assignments.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your assignments.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

Project Format

Students must follow the following requirements when writing and submitting the project report:

- Use Arial, font size 12.
- Include the completed Assessment/Project Coversheet (available on *myLMS*).
- Include a title page, a table of contents, and page numbers.
- Ensure any diagrams, screenshots, and PowerPoint presentations fit correctly on the page and are referenced.
- Use the correct or accurate referencing method throughout, and include a bibliography based on the applicable referencing method at the end of the report.
- Write no more than the maximum word limit.
- Check spelling, grammar, and punctuation.
- Run the report through the Turnitin software.
- Print submissions on both sides of the page.

Essential Embedded Knowledge and Skills Required of Students

- Report-writing and referencing skills
- Ability to analyse scenarios/case studies
- Understanding of subject field concepts and definitions
- Ability to apply theoretical knowledge to propose solutions to real-world problems

Resource Requirements

- A device with Internet access for research
- A desktop or personal computer for typing
- Access to a library or resource centre
- Prescribed reading resources

Delivery Requirements (evidence to be presented by students)

- A typed assignment^[1]
- A Turnitin Originality Report

3. Section A

Section A

Learning Objective

The focus of this project is on the essential web development technologies and tools that can be used to build real-life (industrial-strength) web applications. Students are expected to showcase or demonstrate an understanding of these key web development technologies and tools by building or creating a dynamic web-based application. The project deals with topics like HTML and CSS to create web content and styling, responsive web design, database management, user authentication, and the creation of interactive elements (on web pages) using JavaScript.

Project Topic

Dynamic Web-Based Application Development

Scope

In this project, students are required to select a web application and develop a dynamic web-based application using, HTML, CSS, JavaScript, and the MySQL database. Finally, students must complete and submit the project using the provided template to the lecturer on or before the deadline date.

Marking Criteria

The project must be submitted on or before the due date to the lecturer as per arrangement (or as otherwise stipulated). Five percent (5%) will be deducted for every day that the project is late, up to a maximum of three days. Projects that are more than three days late will be awarded a zero.

3.1. Milestone 1

Milestone 1

25 Marks

a. Students are required to select from the following list a web-based application to develop:

1. Real Estate Website.

- Look at the following example websites: [Property for Sale: Houses for sale: Property24](#)

2. Restaurant.

- Look at the following example website: [Hussar Grill - Best Steak Restaurant | Home](#)

3. A Veterinary Office.

- Look at the following example website: [About Us - Teva Veterinary Clinic.](#)

4. A Book Store.

- Look at the following example website: [Exclusive Books – Exclusive Books Online](#)

5. A Gym

- Look at the following example website: <https://www.planetfitness.co.za/>

b. Perform the following activities:

1. Formulate and describe the project scope of work and all related project details according to the rubric provided.

2. Design the following diagrams for your system

- Wireframes
- Use Case Diagrams
- Class Responsibility Collaborator (CRC) Cards,
- Context Diagram,
- Data Flow Diagram,
- Database Design (showing the data to be stored and the relationship between different elements)
- Class Diagram showing the design patterns you will use in your application

The snippet details or information presented in the Appendix A section provide or serve as extra guidelines for students to elevate their understanding of web development and help them when deciding what they want to build.

Rubric

Criterion (Student must submit)	Mark	Excellent	Good	Developing	Poor	Mark & Comments

1. Wireframes for key pages (home, listing/detail pages, contact/booking etc.)	20	16–20 All key pages included; layout clear and consistent; shows navigation, content blocks and calls-to-action; uses appropriate annotations (labels for sections, buttons, forms); strongly aligned with chosen scenario.	11–15 Most key pages present; layout mostly clear; minor omissions or inconsistencies; alignment with scenario is generally good.	6–10 Some important pages missing or very roughly sketched; little or no annotation; navigation unclear; weak link to scenario.	0–5 Wireframes missing or largely incomplete; pages cannot be interpreted by a developer.	
2. Use Case Diagram (main actors and system use cases)	20	16–20 All relevant actors and major use cases correctly identified; clear system boundary; correct UML notation; relationships (include/extend) used appropriately; diagram readable and logically organised.	11–15 Most actors and use cases included; minor notation mistakes or missing relationships; overall meaning still clear.	6–10 Key actors or use cases missing; major notation issues; relationships unclear; limited usefulness for analysis.	0–5 Diagram absent or incorrect to the point that system behaviour cannot be understood.	
3. CRC (Class Responsibility Collaborator) Cards	10	9–10 CRC cards provided for all key domain classes; responsibilities and collaborators clearly and correctly stated; supports later class diagram design.	7–8 CRC cards for most important classes; some responsibilities vague or overlapping; still usable for design.	3–5 Few CRC cards; responsibilities mostly generic (e.g. "handle data"); collaborators unclear.	0–2 No CRC cards, or cards not related to the chosen system.	
4. Context Diagram (system and external entities)	10	9–10 Context diagram shows the system as a single process with all relevant external entities and data flows named clearly; no internal processes; notation consistent.	7–8 All major entities included; some flows missing or vaguely named; overall system boundary still understandable.	3–5 Several important entities/flows missing; some confusion between internal and external processes.	0–2 Diagram missing or fundamentally incorrect.	
5. Data Flow Diagram (Level 0 or Level 1)	10	9–10 DFD correctly decomposes the system into logical processes, data stores and flows; balance with context diagram; clear labelling; follows DFD rules.	7–8 Mostly correct processes and flows; minor balancing or labelling issues; still useful for understanding data movement.	3–5 Significant gaps or errors (missing stores/flows, mixed physical and logical); difficult to follow.	0–2 No DFD or largely incorrect.	

6. Database Design (ERD / relational schema)	20	16–20 All key entities and attributes included; correct primary and foreign keys; relationships and cardinalities shown; design supports the stated requirements and avoids obvious redundancy.	11–15 Most needed tables/fields present; minor normalisation or key issues; generally supports application.	6–10 Important entities/fields or relationships missing; key definitions unclear; design would be difficult to implement reliably.	0–5 No database design or unrelated tables.	
7. Class Diagram (including chosen design patterns, e.g. MVC, Repository, etc.)	10	9–10 Class diagram consistent with CRC cards and ERD; shows key classes, attributes, methods and relationships; indicates use of identified design patterns (e.g. controller/service/repository separation) clearly.	7–8 Main classes and relationships present; some missing or incorrectly shown; general design intent visible.	3–5 Very few classes; relationships or visibility unspecified; little alignment to CRC/database.	0–2 No class diagram or not meaningful.	

End of Milestone 1

3.2. Milestone 2

Milestone 2: Application Development

75 Marks

1. Develop a dynamic web-based application to meet the project requirements and goals mentioned in Milestone 1. Your web application must be compatible with or adaptable to diverse devices and screens (such as smart phones, tablets, and desktops). Develop the web application using the following technologies

- HTML
- CSS
- JavaScript
- MySQL

2. Add the link to your website on the documentation. Your web application must be compatible with or adaptable to diverse devices and screens (such as smart phones, tablets, and desktops).

3. Note: You must complete and submit your application and project (report) using the provided template on or before the project due date. You will be required to present the application to your lecturer.

Rubric

Criterion	Max - Marks	Excellent	Good	Developing	Poor	Mark & Comments
1. Sample HTML code (structure of key pages; semantic use of tags; forms, navigation)	15	13–15 Clean, valid HTML; correct document structure; good use of semantic tags (header, nav, section, article, footer); forms and inputs correctly defined; code well-indented and readable.	10–12 Overall correct structure; some minor semantic or validation issues; code mostly readable.	5–9 Heavy reliance on non-semantic tags or mis-nested elements; structure works but would be hard to maintain.	0–4 Very little valid HTML; pages largely broken or copied with minimal adaptation.	
2. Sample CSS code (layout, styling, responsiveness)	20	16–20 Uses external stylesheet(s); consistent colour scheme and typography; effective layout with Flexbox/Grid; clear responsive behaviour for mobile, tablet and desktop; CSS organised and commented.	11–15 Good styling and some responsiveness; minor layout or consistency issues; limited comments/organisation.	6–10 Basic styling only; limited or no responsiveness; repeated inline styles or poor structure.	0–5 Minimal or no CSS; application mostly unstyled or broken.	
3. Sample JavaScript code (interactivity, validation, DOM manipulation, use of events)	20	16–20 JavaScript adds meaningful interactivity (e.g. form validation, dynamic content, filtering/search, simple animations); code logically structured and commented; no obvious run-time errors.	11–15 Some interactivity implemented; minor logical or run-time issues; code still understandable.	6–10 Very limited or repetitive scripts; mostly copied examples with little adaptation; some functions not working as intended.	0–5 Little or no JavaScript; or code does not run at all.	

4. MySQL Database and Table Screenshots (evidence of implementation of Milestone 1 design)	20	16–20 Database created in MySQL; tables and relationships closely match the ERD; clear screenshots of structure and sample data; foreign keys and data types appropriate.	11–15 Main tables implemented; some differences from design; screenshots show structure but relationships or data types may be sub-optimal.	6–10 Few tables created; limited alignment with ERD; missing sample data; unclear screenshots.	0–5 No working MySQL implementation or no evidence submitted.	
5. Web Application Functionality (end-to-end behaviour, integration with DB, navigation, responsiveness)	20	16–20 Application meets project requirements for chosen scenario; navigation intuitive; CRUD or relevant core features work correctly with MySQL; responsive on different screen sizes; no critical errors during normal use.	11–15 Most key features implemented and working; some minor bugs or missing secondary functionality; navigation mostly clear.	6–10 Application partially works but has major missing features or errors (e.g. cannot save data, broken links, poor responsiveness).	0–5 Application does not run or fails to demonstrate required functionality.	
6. User Manual (how to access and use the system)	5	5 Clear, concise user manual; includes URL / access instructions, login details (if any), explanation of main features and step-by-step usage with screenshots; written in correct academic language.	3–4 Manual present and understandable; some steps or screenshots missing, but a user can still operate the system.	1–2 Very brief; only high-level notes; many steps unclear; assumes too much prior knowledge.	0 No user manual submitted.	

End of Question 2

4. Template Milestone 1

Template Milestone 1

[Download Word Template](#)

Project Deliverable 1 – Front End Development

Faculty Name:	Information Technology
Module Code:	ITTFA4-14
Module Name:	Front-end and Backend Technologies and Frameworks
Content Writer:	Innocent Manganyi
Copy Editor:	Mr Kyle Keens
Submission Date:	First Block Week 6

Student Name	
Student Number	
Project Title	
Submission Date	

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1.1 Documentation

Provide the following diagrams for your web application.

- Wireframe showing the interface design of your web application (This might not necessary be exhaustive of all application pages, but it should reflect the key pages of your web application).
- Use Case Diagrams showing the use cases of your web application. This might not necessary be exhaustive of all use cases, but it should reflect the use cases of your web application
- Class Responsibility Collaborator (CRC) Cards,
- Context Diagram,
- Data Flow Diagram (showing the data to be stored and the relationship between different elements)
- Database Design
- Class Diagram showing the design patterns you will use in your application

5. Template 2 Milestone 2

Template Milestone 2

[Download Template](#)

Project Deliverable 2 – Back End Development

Faculty Name:	Information Technology
Module Code:	ITTFA4-14
Module Name:	Front-end and Backend Technologies and Frameworks
Content Writer:	Innocent Manganyi
Copy Editor:	Mr Kyle Keens
Submission Date:	Forth Block Week 6

Student Name	
Student Number	
Project Title	
Submission Date	

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1.1 Coding

Provide the screenshots and code samples of the final website. You do not have to print every code on your website. Use the following as guideline and explain what each sample code provided is used for on the website:

- Sample HTML Code
- Sample CSS Code

- c. Sample JavaScript Code
- d. Sample MySQL code

1.2 Web Application Front-End Functionality

Upload a zipped folder of your application. Present your application

1.3 Web Application Back End Functionality

Upload a zipped folder of your application. Present your application.

1.4 User Manual

Write a manual indicating the functionalities for each user for the system. Upload the user manual.

6. Appendix

Appendix A

Systems Requirements

1. Requirements for Developing a Real Estate Website

User Roles:

- **Guest Users:** Browse properties, search listings, and view limited details.
- **Real Estate Agents/Sellers:** List properties, manage listings, and track inquiries.
- **Admin:** Manage website content, users, and system settings.

Core Features:

1. Property Listings:

- o Property details: images, price, description, location, amenities, and availability.
- o Advanced search and filtering: by price, location, property type, number of bedrooms, etc.

2. Property Posting (For Sellers/Agents):

- o Upload property details and images.
- o Manage property status (available, sold, etc.).

3. Property Search:

- o Keyword-based search with autocomplete suggestions.
- o Map-based search to locate properties.
- o Filters for advanced property searches.

4. Communication:

- o Inquiry forms to contact agents/sellers.

5. **Featured Properties:**

- o Showcase premium or highlighted properties on the homepage.

6. **Property Comparisons:**

- o Compare properties based on price, size, amenities, and more.

Optional Advanced Features:

- Mortgage calculator for buyers

2. Requirements for Developing an Online Restaurant Website:

1. **Homepage:**

- o Highlight restaurant's branding (logo, tagline, and featured images).
- o Carousel or banner for promotions, featured dishes, or events.

2. **Menu Section:**

- o Display dishes with images, descriptions, and pricing.
- o Categorize items (e.g., Starters, Steaks, Desserts, Beverages).
- o Option to filter or sort (e.g., vegetarian, gluten-free).

3. **Reservation System:**

- o Date, time, and party size input.
- o Real-time availability check.
- o Confirmation via email/SMS.

4. **About Us:**

- o Restaurant history, mission, and values.
- o Chef profiles and team introductions.
- o Awards or recognitions.

5. **Locations & Contact:**

- o List of restaurant branches with addresses, contact numbers, and hours.
- o Contact form for inquiries (with CAPTCHA for spam protection).

6. **Online Ordering System (Optional):**

- o View menu items and add to cart.
- o Checkout with delivery or pickup options.
- o Payment gateway integration (credit cards, mobile wallets).

7. **Reviews and Testimonials:**

- o Display customer feedback.
- o Integration with platforms like Google Reviews or TripAdvisor.

8. **Social Media Integration:**

- o Links to platforms like Instagram, Facebook, and Twitter.
- o Embedded Instagram feed to showcase latest posts.

3. Requirements for Developing a Veterinary Clinic Website:

1. Homepage:

1. Welcoming introduction with clinic tagline and featured images of the facility and pets.
2. Call-to-action (CTA) buttons for appointments, services, and contact information.
3. Highlighted sections for promotions, testimonials, or news.

2. About Us Section:

1. Detailed information about the clinic's history, mission, and values.
2. Profiles of veterinarians and staff, including qualifications and specialties.
3. Awards, recognitions, or certifications.

3. Services:

1. List of veterinary services offered (e.g., wellness exams, vaccinations, surgical procedures).
2. Service descriptions with FAQs for each.
3. Pricing details

4. Online Appointment Booking:

1. Date, time, and pet details input.
2. Confirmation via email/SMS.
3. Option to reschedule or cancel appointments.
4. Payment of services

5. Shopping Cart:

1. Add, remove, or update booking
2. Display total price.

6. Secure Checkout:

1. Multiple payment methods (credit card, PayPal, etc.).
2. Booking confirmation via email.

5. Pet Care Resources:

1. Blog articles on pet health, nutrition, and care tips.
2. Downloadable resources (e.g., vaccination schedules, diet plans).

6. Contact Us Section:

1. Clinic address, phone number, and email.
2. Contact form for inquiries (with CAPTCHA for spam prevention).

7. Emergency Services:

1. Highlight emergency services with a dedicated page or section.
2. Emergency contact details prominently displayed.

8. Testimonials and Reviews:

1. Display feedback from pet owners.
2. Integration with Google Reviews or similar platforms.

4. Requirements for Developing a Bookstore Website: *Exclusive Books*

User Roles:

1. Users:

- o Browse books and categories.
- o View book details and reviews.
- o Create and manage user profiles.
- o Save books to Wishlist or shopping cart.
- o Purchase books

2. Admin:

- o Manage book inventory, orders, and customer queries.
- o Update promotions and featured books.

Core Features:

1. Homepage:

- o Highlight new arrivals, bestsellers, and promotions.
- o Search bar with autocomplete for quick book search.
- o Category navigation (e.g., Fiction, Non-fiction, Children's Books).

2. Book Catalogue:

- o Filter and sort books by genre, price, author, and rating.
- o Display book details, including title, author, price, reviews, and availability.
- o High-quality images and preview options (e.g., excerpt or table of contents).

3. Book Details Page:

- o Detailed description of the book (synopsis, genre, and ISBN).
- o Author biography and related works.
- o Customer reviews and ratings.

4. Shopping Cart:

- o Add, remove, or update book quantities.
- o Display subtotal, shipping costs, and total price.

5. Secure Checkout:

- o Multiple payment methods (credit card, PayPal, etc.).
- o Order confirmation via email.

6. Search Functionality:

- o Keyword-based search with advanced filters (e.g., genre, publication year).
- o Autocomplete suggestions for titles, authors, or categories.

7. Admin Panel:

- o Add, update, or remove books and categories.
- o Manage user accounts, orders, and inventory.

- o Generate sales and customer analytics reports.

8. **Promotions and Discounts:**

- o Highlight special offers or bundle deals.
- o Coupon codes for discounts at checkout.

9. **Contact Us:**

- o Customer support contact form or live chat.
- o FAQ section for common queries (e.g., returns, shipping).

5. **Requirements For for Developing a Gym Website**

1. **Home Page:**

- o Display the brand logo, tagline, and a brief introduction.
- o Highlight key promotional offers, new programs, and membership deals.
- o Include links to social media platforms and contact details.

2. **Membership Plans:**

- o Provide details about various membership options (e.g., monthly, yearly, family plans).
- o Allow users to compare plans.
- o Include an online sign-up and payment system.

3. **Gym Locations:**

- o An interactive map with all Gym branches in South Africa.
- o Search functionality based on city, area, or postal code.
- o Show contact details, operating hours, and amenities for each location.

4. **Classes and Programs:**

- o List available fitness classes (e.g., yoga, Zumba, HIIT).
- o Display schedules for each class and location.
- o Allow users to book classes online.

5. **Personal Training:**

- o Provide information about personal trainers.
- o Allow users to book personal training sessions.

6. **Health and Wellness Blog:**

- o Regular articles on fitness tips, nutrition, mental wellness, and lifestyle.
- o Option for users to subscribe to the newsletter.

7. **Contact and Support:**

- o Contact form for inquiries and feedback.
- o FAQ section addressing common questions.
- o Live chat or chatbot feature for immediate assistance.

8. **Promotions and Offers:**

- o Dedicated page for current and upcoming promotions.
- o Option to download promotional vouchers or codes.

9. Mobile Responsiveness:

- o Ensure the website is fully functional and aesthetically pleasing on mobile devices.

10. Social Media Integration:

- o Include links and live feeds from Planet Fitness South Africa's social media accounts.
- o Encourage users to share their fitness journey with hashtags.

11. Privacy and Security:

- o Adherence to South African data protection laws.
- o SSL certification for secure transactions.
- o Clear privacy policy and terms of service.

Web Application(s) Types - Examples of Web Apps + Benefits and Challenges

Web application types will influence any part of your project, from the technology stack to your web application's functionality, performance, speed of development, and scalability.

The range is extensive, from static web applications – known for their simplicity and ease of maintenance – to dynamic web applications that bring content to life with real-time updates. Then, there are single-page applications, offering a seamless user experience with minimal loading time. Progressive web applications stand out, too, working offline and feeling like a native app on your device.

In this blog, we'll provide a clear, straightforward guide exploring 11 web application types in detail and help you make informed decisions for your digital endeavours.

What is a Web Application?

A web application is a software program that runs directly in a web browser. Unlike traditional desktop applications, it doesn't require downloads or installations.

Many types of web applications are built using client-side scripts (like HTML, JavaScript, or CSS) and server-side scripts (like PHP or ASP). They interact with users by sending requests to the server, which processes the data and returns the results. This setup allows web applications to perform various functions, from displaying content to managing user data.

They're accessible from any device with an internet connection, making them versatile for many uses, from personal blogs to complex e-commerce sites. Web app development services typically follow a meticulous process in creating a custom web application, involving careful planning, design, coding, testing, and maintenance.

11 Types of Web Application Development

Here are 11 categories of web applications, each with its unique features and use cases.

1. Static Web Applications

These are the simplest types of web applications built using HTML and CSS, suitable for creating portfolios or digital brochures. As the name suggests, the content only changes if manually updated by the developer. Static web applications are straightforward to create and host, as they don't require extensive server-side processing.

This option is a cost-effective solution for individuals or small businesses needing a simple online presence. However, their simplicity also means limited functionality.

2. Dynamic Web Applications

In contrast, dynamic web applications are more complex and interactive. They use client-side and server-side scripts (like JavaScript, PHP, ASP, or JSP) to generate content in real-time. These web application categories are connected to a database, allowing them to provide personalized experiences based on user interactions and preferences.

They're ideal for businesses, especially if your top priorities are user engagement and content variability. Due to their complexity, dynamic web applications are more challenging to develop and maintain. They require a more robust hosting environment and higher web development costs.

3. Single-Page Applications (SPAs)

SPAs load a single HTML page and dynamically update content as users interact with the app. These categories of web applications are ideal for platforms where user experience and speed are critical, such as:

- Social media platforms
- Email clients
- Cloud-based software

The benefit is that this web app type avoids reloading the entire page with each user action, leading to a smoother and faster user experience. However, they also come with challenges, particularly in SEO optimization and initial load times, as the entire application must be loaded simultaneously.

SPAs are built using JavaScript frameworks like Angular, React, or Vue.js, which handle the dynamic loading of content and user interface elements.

4. Multi-Page Web Applications (MPAs)

Unlike SPAs, MPAs reload the entire page from the server when the user interacts with the application.

These traditional web application categories are more suitable to websites with a large amount of content and diverse functionalities, such as:

- eCommerce sites
- Online catalogs
- Educational platforms

MPAs can handle complex structures and vast databases more efficiently than SPAs. They're also better optimized for search engines, as each page can be indexed separately. However, MPAs often have slower page transitions and can be more resource-intensive, as each new page needs a server request and page reload.

Developing MPAs typically involves a more extensive back-end process to manage multiple pages and their interactions with the server. When considering developing these web application types, you can partner with a skilled website development company to ensure a seamless and effective online presence.

5. Progressive Web Applications (PWAs)

PWAs represent a hybrid of regular web pages (or websites) and a mobile application. They're installable on a device's home screen without downloading from an app store. A key feature of PWAs is the use of service workers or scripts running in the background, enabling:

- Offline functionality
- Push notifications
- Background data syncing

PWAs are also responsive and linkable, which can be shared via a URL. They offer a high level of performance, engaging users with smooth animations and no janky scrolling. Thus, these categories of web applications are incredibly efficient, especially for users with limited internet connectivity. Developers use standard web technologies to build them, including HTML, CSS, and JavaScript.

6. Content Management Systems (CMS)

A CMS manages creation and modification of digital content, supporting multiple users in a collaborative environment. CMS features vary widely, including but not limited to:

- Web-based publishing
- Format management
- History editing
- Version control
- Indexing
- Search

They're suitable for blogging, e-commerce, and news websites, where you need to frequently update the content without extensive technical know-how. CMS platforms like WordPress, Joomla, and Drupal are popular choices, offering templates and plugins for customization without needing to write code from scratch. What's more, CMSs provide a user-friendly interface, allowing for easy updates and managing content.

7. eCommerce Web Applications

These web application types facilitate online buying and selling. They're complex systems that integrate various functionalities, including:

- Product displays or catalogs
- Product search and filtering
- Shopping carts
- Payment processing
- Customer account and order management
- Customer service tools

E-commerce web applications must provide a seamless, user-friendly experience to encourage sales and repeat business. These applications must be scalable to handle varying traffic and sales volume levels. Also, security is paramount to protect sensitive customer data, including payment

information.

Platforms like Shopify, Magento, and WooCommerce are famous examples, offering customizable templates and various plugins to enhance functionality. E-commerce web applications have revolutionized the retail industry, allowing businesses to reach a wider audience and operate 24/7.

8. JavaScript-Powered Web Applications

JavaScript is a versatile programming language to create dynamic and interactive user experiences. It's the most used programming language in 2023. JavaScript can be used both on the client side (in the browser) and the server side (with technologies like Node.js), making it a powerful tool for full-stack development.

JavaScript-powered web applications are known for their speed and efficiency; they can update content without reloading the entire page. This functionality makes them ideal for applications that require real-time data updates, such as:

- Social media platforms
- Online games
- Collaboration tools

React, Angular, and Vue.js are examples of JavaScript frameworks and libraries. They provide pre-written JavaScript code to handle everyday tasks.

9. Rich Internet Web Applications (RIAs)

RIAs are advanced web application types that deliver a user experience like desktop applications. They use client-side frameworks to provide interactive features and a richer user interface, such as:

- Adobe Flash
- JavaFX
- Microsoft Silverlight

RIAs run inside web browsers but behave like desktop applications, offering responsive, engaging user experiences with better data visualization and real-time interaction capabilities. RIAs can process data and perform tasks without constantly communicating with the server, reducing load times and improving performance. However, they require plugins or specific frameworks, which might limit accessibility and compatibility across different devices and browsers.

10. Portal Web Applications

Portal web applications are gateways to various information, services, and other applications. Enterprises often use them for internal purposes or to provide customer-facing services, for example: These web application types typically require user authentication, offering personalized content and a centralized access point to various resources.

Portal web applications are designed to aggregate content from different sources, providing a consistent and integrated user experience. They can handle various functionalities, including:

- Search engines
- Email systems
- Forums
- Newsfeeds

11. Animated Web Applications

These applications focus on delivering rich visual content and interactive elements using animations. They're particularly popular in fields that need high levels of user engagement, such as:

- Online advertising
- Gaming
- Educational platforms

Animated web apps are built using technologies like CSS3, HTML5, and WebGL, so developers can create complex yet engaging user interfaces. They provide dynamic and visually appealing experiences, capturing users' attention and improving interaction. However, developing these applications can be time-consuming and require advanced design as well as programming skills.

Furthermore, you need to balance the animations to enhance rather than hinder the user experience, especially considering performance and accessibility on various devices.

Extracted from Source: Bhatt, T. 2023. Web Applications Types - Examples Of Web Apps + Benefits And Challenges [Online]. [23 January 2024]